

S. cerevisiae Culture

This protocol outlines the general steps for preparing an *S. cerevisiae* culture for transformation or otherwise, measuring OD600 to create a growth curve, determining transformation conditions, in addition to creating glycerol stocks from the resulting cultures.

This protocol also includes details for the culture of experimental media, though the instructions for the growth curve remain general.

(Glycerol stock → Agar) about 2 days

(Starter liquid culture from colony) about 6-8h incubation / overnight

Expect growth curve to also take 6-8h

Reagents

✓	Name	Quantity	Location/Notes
☐	Glycerol stock of <i>S. cerevisiae</i>	1	Freezer (Day 1)
☐	YPD agar plate	1	For glycerol stock (Day 1)
☐	YPD Media (if not transformed) SD Media (if transformed)	4mL	For starter culture (Day 3, leave overnight)
☐	Experimental media	As needed, 25mL each	For growth curve (Day 4)
☐	30% glycerol	500 μ L per culture	For storage (any stationary phase)

Equipment

✓	Name	Quantity	Location/Notes
☐	Incubator (30°C)	1	Standard yeast growth
☐	Shaking incubator (~200 rpm)	1	Liquid cultures
☐	Culture tubes (14 mL)	As needed	For starter cultures
☐	Sterile inoculating loop	As needed	For colony transfer
☐	Petri dishes	As needed	Agar plates
☐	Vortex mixer	1	Resuspending cultures
☐	Micropipettes / sterile tips	As needed	Liquid handling
☐	-80°C freezer	1	Glycerol stock storage
☐	Culture flasks (≥50mL)	As needed	≤50% full for aeration
☐	Disposable cuvettes	As needed	OD readings
☐	Sterile hood	1	Sterile technique
☐	Spectrophotometer (OD600)	1	Growth measurement

Instructions

PART A: PREPARE STARTER CULTURE

NOTE: step 1 is only necessary if starting from glycerol stock – if starting from isolated colony already on plate, skip to step 2

1. Prepare or obtain yeast colonies on solid media (for starting liquid culture)

- Use a sterile inoculating loop to pick a small amount of yeast cells (thin smear on loop – not a clump) from an existing plate or glycerol stock

- Streak the cells onto a fresh YPD agar plate (thin out to obtain isolated colonies)
- Invert the plate and incubate at 25°C or 30°C
- Allow colonies to grow until they are approximately 1 mm in diameter
 - Takes ~2 days for wild-type strains on rich media

Reagent Tracker (use this table as needed)	Quantity
☐ YPD agar plate	1
☐ Yeast strain (from plate or glycerol stock)	Small amount

2. Prepare a starter culture (pre-culture, liquid culture)

- Pick a single, well-isolated colony from an SD agar plate (post-selection)
- Inoculate the colony into 4 mL of sterile SD broth in a culture tube
- Grow the culture with shaking at ~200 rpm at 30°C to ensure proper aeration.
- Incubate for 6–8 hours or overnight, until cells are actively growing.
- This starter culture will be used to inoculate all experimental conditions

Reagent Tracker (use this table as needed)	Quantity
☐ Yeast colony	1
☐ YPD or SD media	4 mL

PART B: GROWTH CURVE WITH EXPERIMENTAL MEDIA

This experiment compares yeast growth across control and fatty acid-supplemented conditions to assess the effects of long-chain and short-chain fatty acids on growth dynamics.

3. Prepare control and experimental cultures

- Mix the starter culture thoroughly to ensure cells are evenly suspended (pipette up and down).
- The starter culture will serve as the inoculum for all treatment conditions.

- Dispense 25 mL of each prepared medium into separate sterile culture vessels:
 - Control
 - Experimental 1
 - Experimental 2
 - Experimental 3
- Inoculate at an approximate dilution of 1:100 to 1:1000 (25–250 μ L starter culture into 25 mL media).
 - Initial OD600 of approximately 0.2–0.3.

Reagent Tracker (use this table as needed)	Quantity
☐ Prepared Media	25 mL per condition
☐ 25–250 μ L starter culture	25–250 μ L per condition

4. Grow cells to mid-log phase and monitor growth (OD600 $\rightarrow$ cell density)

- Incubate cultures at 30°C with shaking (~200 rpm) under aerobic conditions.
- At regular intervals (every ~2 hours), remove 1 mL of culture from each condition.
- Prepare a blank using 1 mL of the corresponding uninoculated medium.
- Measure the optical density at 600 nm (OD600) for each sample.
- Record OD600 values and note any visible differences in:
 - Turbidity
 - Colour
 - cell settling
- Continue sampling until cultures reach mid-exponential (log) phase, defined as:
 - OD600 $\approx$ 0.4–0.9
 - Optimal range: 0.5–0.7
- If OD600 exceeds 1.0, dilute the sample (e.g., 1:10) with the same medium and re-measure.

Reagent Tracker (use this table as needed)	Quantity
☐ Yeast culture (x4)	~1 mL per reading
☐ Blank media (x4)	~1 mL per reading

5. Harvest cells (mid-log phase)

- Transfer an appropriate volume of culture (e.g., 5–10 mL) into a sterile centrifuge tube.
 - Do this for each of the experimental conditions (x4)
- Centrifuge the culture at $\sim 3000\text{--}5000 \times g$ for 3–5 minutes to pellet the cells.
- Carefully remove and discard the supernatant without disturbing the cell pellet.
- Resuspend the cell pellet in sterile distilled water or appropriate transformation buffer
- Proceed immediately with the transformation protocol.

Reagent Tracker (use this table as needed)	Quantity
☐ Yeast culture (x4)	Varies

PART C: STORING FOR LONG-TERM USE

6. Store strains for long-term use (optional??)

- Grow a yeast culture to stationary phase
- Mix 500 μL of culture with 500 μL of sterile 30% glycerol
- Label tube clearly
- Store at -80°C
- Do not allow glycerol stocks to thaw and refreeze

Reagent Tracker (use this table as needed)	Quantity
☐ Yeast culture	500 μL
☐ Sterile 30% glycerol	500 μL

Notes

- 30°C is the standard growth temperature for budding yeast; 25°C is also commonly used
- Growth phases (similar to *E. coli*):
 - Lag phase:
 - Cells adjust to new environment
 - Growth is slow
 - This phase can last several hours
 - Log phase (exponential growth):
 - Cells grow at maximum rate → best phase
 - Mid-log: OD600: 0.4-0.9 → 0.5-0.7 ideal
 - Gene expression highest
 - IF OD > 0.7 but still < 2.0:
 - Dilute to ~0.3 → regrow to 0.6
 - Doubling time:
 - Healthy cells (rich media, 30°C): ≤ 90 minutes
 - Mutants: can be several hours
 - Late log phase:
 - Nutrients (especially sugar) becomes limiting
 - Growth slows
 - OD600: 0.9-2.0
 - Acceptable for:
 - Large biomass collection
 - Less sensitive experiments
 - Stationary phase:
 - Nutrients depleted → cells stop dividing but remain viable
 - OD600 > 2
 - Not ideal for most experiments
 - Expectations: DNA extraction
 - If transferred to fresh media: cells resume growth (after lag phase)

- Types of media:
 - Rich media
 - Faster growth
 - More variability
 - Synthetic media (SD/SC)
 - Slower growth
 - Precise control of nutrients
 - Required for:
 - Selection (auxotrophic markers)
 - Maintaining plasmids
 - Would use this after transformation
- Media naming:
 - Rich media:
 - YP + carbon source → example: YPD = glucose
 - Synthetic media: S + carbon source → example: SD/ SC
 - Supplements indicated by -U, -L, -H
- Sterility & Contamination
 - Maintain sterile technique (e.g., flame or sterile hood)
 - Discard cultures if:
 - yeast cells should settle at the bottom; if not, contamination may be present
- Initial colonies can all be streaked on agar plates but should be grown under experimental conditions



References

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